

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 5

3 Computer Networking (awm22)

Ethernet,
switches,
topology
discovery using
spanning tree.

- (a) (i) Ethernet switches use a spanning tree. Explain briefly what the aims of the spanning tree protocol are, and how it works to achieve these aims.

[6 marks]

Answer: salient points

A tree whose crucial property
Is loop-free connectivity.

A tree which must be sure to span
So packets can reach every LAN.

First the root must be selected.
By ID it is elected.

Least cost paths from root are traced.
In the tree these paths are placed.

A mesh is made by folks like me
Then bridges find a spanning tree.

Radia Perlman 1985

- (ii) Do the switches learn the entire network topology? Explain your answer.

[2 marks]

Answer: No. The switches do not learn the entire topology – only the distances to the root from each neighbour.

- (iii) Routers using link-state protocols communicate over the shortest path. Does each pair of switches communicate over the shortest path?

[2 marks]

Answer: No. Switches communicate following a single spanning tree. The tree is rooted at the switch with the smallest id. As a result some links are never used. As a result traffic may not follow the shortest path.

- (iv) Routers may use a link-state protocol or a distance vector protocol. Compare the message-size complexity, computational complexity, and the robustness of these two approaches.

[6 marks]

Answer:

Message complexity

LS: $O(N \times E)$ messages;

N is #nodes; E is #edges

DV: $O(\text{\#Iterations} \times E)$

where #Iterations is ideally $O(\text{network diameter})$ but varies due to routing loops or the count-to-infinity problem

Processing complexity

LS: $O(N^2)$

DV: $O(\text{\#Iterations} \times N)$

Robustness: what happens if a router malfunctions?

LS:

node can advertise incorrect link cost

each node computes only its own table

DV:

node can advertise incorrect path cost

each node's table used by others; error propagates through network

IPv4 IPv6 -
zeroconf and
DNS question

- (b) The computer of another student on your college stairwell doesn't connect to the Internet.

They go on to say "...it's weird — my computer can talk with the printer in my room and my computer can see your computer, but I can't upload this essay due tonight, I can't connect to Google, and I can't even send an email..."

You suspect their computer is using automatically-allocated *link-local* addresses for both IPv4 and IPv6.

Speculate what has gone wrong to cause the computers to be using link-local addresses. You may want to consider: What address did the machine use? Why is the computer using link-local addresses? Why are some services are working but other services are not?

[4 marks]

Answer:

The computer does not have a routable IP address. Link-local addresses will only communicate with other devices on the same *link* - that is - devices that can communicate directly with each other (no router required.)

Same wifi network, same ethernet switch, etc.

IPv4 link-local addresses are assigned from address block 169.254.0.0/16 (169.254.0.0 through 169.254.255.255).

In IPv6, they are assigned from the block fe80::/10

The other hosts that work are also (re)configured to use link-local addresses. This is because link-local addresses can be allocated without a server and may simply be because of a protracted issue such as those below

Another alternative is that the IPv6 was always link-local but it was only the IPv4 (DHCPv4) services that had failed and hence link-local computers with IPv6 continued working happily.

Short sentence answers expected such as

DHCP server broken

or

local router is disconnected

— *Solution notes* —

or

access point is isolated

The answer is (typically) that the machine can't complete/receive DHCP and is relying on link-local allocations instead.
